WASP-77Ab

R-1267

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-77_220120` · created 2026-07-31T13:22:39+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459599.65063 +/- 0.00055 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.0 +/- 0.013
Transit depth (Rp/Rs)^2	1.1e-07 +/- 8.8e-05 [%]
Orbital Inclination (inc)	88.35 +/- 1.43
Airmass coefficient 1 (a1)	1.198 +/- 0.011
Airmass coefficient 2 (a2)	-0.1288 +/- 0.0068
Scatter in the residuals of the lightcurve fit is	0.91 %
Best Comparison Star	None
Optimal Aperture	5.18
Optimal Annulus	12.51
Transit Duration (day)	0.0793 +/- 0.0024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.0 ± 0.013 vs 0.1301 (deviation 7.39σ)
Duration fit vs published	0.0793 d vs 0.0908 d (deviation 3.55σ)
Mid-time O–C	1.0 min
Depth SNR	0.0
Residual scatter	0.91 %

Light curve

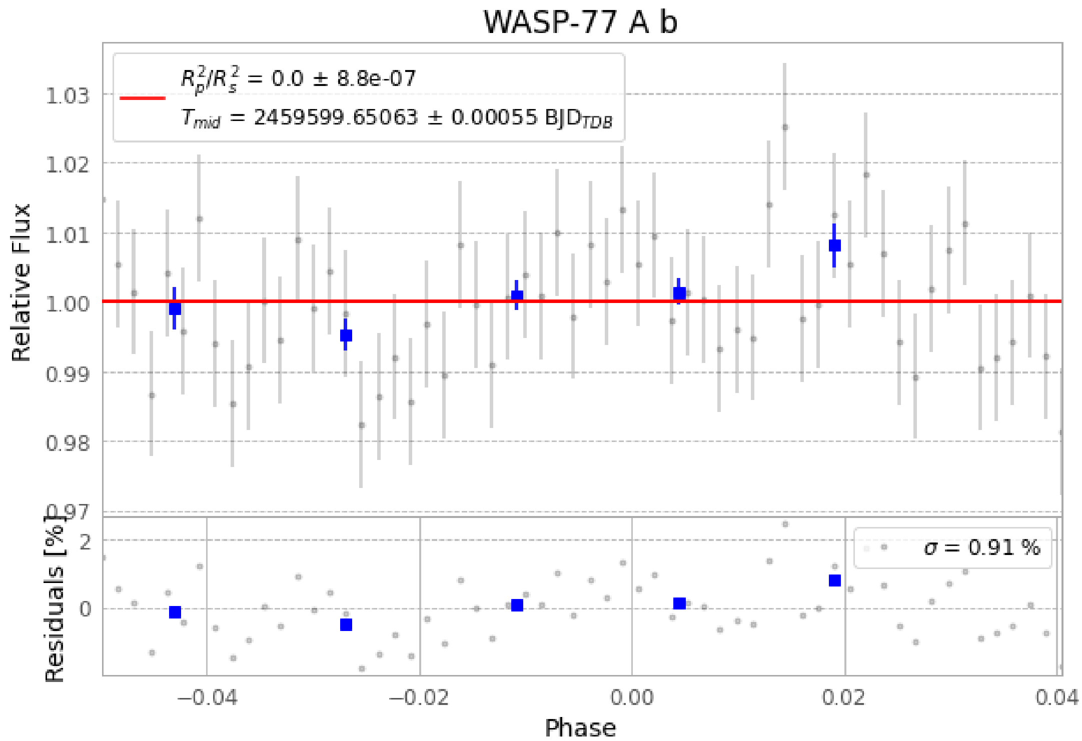

Figure 1 — FinalLightCurve_WASP-77 A b_2022-01-19.png (SHA-256 a2fad2b0cc05c6e0…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [552, 184], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 5ad60c4842c189a6…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

60 FITS frames (39 MB), `WASP-77220120015712.FITS` → `WASP-77220120045413.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-77-220120.log (84a89867ee95…), FinalLightCurve_WASP-77 A b_2022-01-19.png (a2fad2b0cc05…), FinalLightCurve_WASP-77 A b_2022-01-19.csv (e6b803c1b099…)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)